B.Sc. (Hons) Agriculture – 2025 Syllabus

Second Year – First Semester

S. No	Course No	Course Title	Credit Hours
1	GPB 201	Principles of Genetics	3 (2+1)
2	AGR 201	Crop Production Technology-I (Kharif Crops)	3 (1+2)
3	HOR 211	Production Technology of Fruit and Plantation Crops	2 (1+1)
4	AEX 201	Fundamentals of Extension Education	2 (1+1)
5	NEM 201	Fundamentals of Nematology	2 (1+1)
6	AGR 202	Principles and Practices of Natural Farming	2 (1+1)
7	SAC 201	Soil Resource Inventory and Geoinformatics	1 (1+0)
8	AEC 201	Farm Management, Production and Resource Economics	2 (1+1)
9	SWC 211	Fundamentals of soil and water conservation engineering	2 (1+1)
10	SEC V	Skill Enhancement course-V	2(0+2)
11	AGR 203	Study Short Tour	1(0+1)*
12	NSS 201 / NCC 201	National Service Scheme II / National Cadet Corps II	1 (0+1)#
13	PED 101	Physical Education, First Aid, Yoga Practice and Meditation (PED and First aid offered in III semester)	2(0+2) **
		Total	**22(10+12)**

Class Duration:

Practical: 2 hours

Theory: 1 hour

GPB 201 Principles of Genetics (2+1)

Theory

Unit I: History, cytology and chromosomes

Introduction and definition of cytology, genetics, heredity, inheritance and cytogenetics and their interrelation. Brief history of developments in genetics and cytogenetics. Prokaryotic and eukaryotic cells and cell organelles. Cell cycle and cell division-mitosis and meiosis and their significance. Sporogenesis, gametogenesis and syngamy in plants, identical and fraternal twins. Chromosomes, structure and chemical composition, karyotype and ideogram. Types of chromosomes based on structure and function, position and number of centromeres. Special types of chromosomes - polytene, lampbrush, B chromosomes, ring and isochromosomes. Structural variation in chromosome– deletion, duplication, inversion and translocation, genetic and cytological implications.

Unit II: Chromosomal numerical variations, model organisms, Mendelian laws and deviations of Mendelian inheritance

Numerical variations in chromosome - aneuploid, types of aneuploids and their origin; Klinefelter syndrome and Turner syndrome; and their implications. Euploid - haploids, dihaploids and double haploids in genetic.Polyploid - auto and allopolyploids and their characters, evolution of wheat, triticale, cotton, tobacco and *Brassica.* Pre and post mendelian concepts of heredity, vapour and fluid theory, magnetic power theory, preformation theory, Lamarck's theory, Darwin's theory, germplasm theory and mutation theory. Study of model organisms - Drosophila, Arabidopsis, garden pea, E. coli, and mice. Probability and Chi-square. Mendel's experiments and laws of inheritance. Rediscovery of Mendel's work.. Important terminologies. Chromosomal theory of inheritance. Deviation from Mendelian inheritance. Allelic interactions – dominance vs recessive, complete dominance, codominance, incomplete dominance, threshold characters.

Unit III: Qualitative and quantitative traits, lethal genes, multiple alleles and crossing over

Epistatic interactions with example -deviation from Mendelian inheritance – non allelic interaction without modification in Mendelian ratio – Bateson and Punnett's experiment on fowl comb shape. Non allelic interaction with modification in Mendelian ratio – i.) dominant epistasis (12:3:1). ii.) recessive epistasis (9:3:4) iii.) duplicate and additive epistasis (9:6:1). iv.) duplicate dominant epistasis (15:1)duplicate recessive epistasis (9:7) vi.) dominant and recessive epistasis (13:3),summary of epistatic ratios. Qualitative and quantitative traits, polygenes and continuous variations – transgressive segregation, multiple factor hypothesis - Nilsson Ehle experiment on wheat kernel colour, corolla length in *Nicotiana longiflora.* Lethal genes, pleiotropism, penetrance and expressivity, multiple alleles, blood group genetics in humans, coat colour in rabbits, self incompatibility in plants,Pseudoalleles and isoalleles. Crossing over mechanism– significance of crossing over, cytological proof for crossing over - Stern's experiment, factors controlling crossing over.

Unit IV: Linkage, sex determination, mutation and cytoplasmic inheritance

Linkage and its estimation, coupling and repulsion - experiment on Bateson and Punnet. Morgan's chromosomal theory of linkage, complete and incomplete linkage, linkage group. Strength of linkage and recombination, double cross over, interference and coincidence. Two point and three point test cross genetic map, physical map. Sex determination - autosomes and sex chromosomes, chromosomal theory of sex determination, haplodiplod type, sex determination in plants – *Melandrium*, papaya, maize,genic balance theory of Bridges and gynandromorphs.Sex linkage, sex linked inheritance – criss cross inheritance, reciprocal differences, holandric genes, sex limited and sex influenced traits. Mutation - classification, mutagenic agents and mutation induction methods. Maternal effects – pattern of snail shell coiling, Cytoplasmic inheritance– features of cytoplasmic inheritance, chloroplast- plastid colour in *Mirabilis jalapa*, mitochondrial - cytoplasmic male sterility in maize. Infective particles - kappa particles in paramecium.

Unit V: DNA structure and function, central dogma of life

DNA-the genetic material – Griffith's experiment, experiment of Avery, McCleod and McCarty, confirmation by Hershey and Chase. RNA as genetic material – Frankel, Conrat and Singer experiment. Types of DNA and RNA, Watson and Crick model, Models of DNA replication, proof for semi conservative method of DNA replication, steps involved in DNA replication. Genetic code, cistron, muton and recon. Gene concept - Gene structure , molecular concept of gene, molecular definition of a gene, number of genes on a single chromosome, size of a gene, fine structure of a gene, types of genes. Central dogma of life - transcription and translational mechanism of genetic material and protein synthesis,regulation of gene expression – Operon model of Jacob and Monad – Lac and Trp operons.

Practicals

1. Study of microscope. Study of cell structure.Principles of killing and fixing; preparation of stains and preservatives. Study the mitosis -onion / *Aloe sp.*meiosis cell division in rice/maize, peralmillet/sorghum, making permanent slide.Probability and chi-square test. Experiments on monohybrid, dihybrid, trihybrid, test cross and back cross. Experiments on simple interaction of genes-comb character in fowls, dominant epistasis, recessive epistasis, duplicate and additive epistasis, duplicate dominant epistasis, duplicate recessive epistasis, dominant and recessive

epistasis including test cross and back cross. Multiple alleles and polygenic inheritance. Problems on linkage and cross-over analysis (through two point and three point test cross data). Working out interference, coincidence and drawing genetic maps. Problems on sex linked inheritance in Drosophila and Human. Study on models on DNA and RNA structures.

Theory Schedule

1. Introduction and definition of cytology, genetics, heredity, inheritance and cytogenetics and their interaction. Brief history of developments in genetics and cytogenetics.
2. Cell and cell organelles. Differences between prokaryotes and eukaryotes.
3. Cell cycle, cell division-mitosis and meiosis and their significance.
4. Sporogenesis, gametogenesis and syngamy in plants, identical and fraternal twins.
5. Architecture of chromosomes, chemical composition, chromonemata, chromosome matrix, nucleosome, centromere, telomere, euchromatin, heterochromatin, NOR, secondary constriction, satellite chromosome, karyotype, ideogram.
6. Types of chromosomes based on position and number of centromeres, structure and function, special types of chromosomes - polytene, lampbrush, B chromosomes, ring and isochromosomes.
7. Chromosomal aberration: variation in chromosome structure – deletion, duplication, inversion and translocation – genetic and cytological implications.
8. Chromosomal aberration: variation in chromosome number –aneuploid, types of aneuploids and their origin; Klinefelter syndrome and Turner syndrome and their implications.
9. Euploid - haploids, dihaploids and double haploids in genetics . Polyploid - auto and allopolyploids, their characters, evolution of wheat, Triticale, cotton, tobacco, *Brassica.*
10. Pre-Mendelian concepts of heredity – vapour and fluid theory, magnetic power theory, preformation theory, Lamarck's theory, Darwin's theory, germplasm theory and mutation theory.
11. Study of model organisms -Drosophila, Arabidopsis, Garden pea, E. coli, and mice.
12. Probability and chi-square.
13. Mendel's experiments and laws of inheritance. Rediscovery of Mendel's work.
14. Terminologies - gene, allele, locus, homozygous, heterozygous, hemizygous, genotype, phenotype, monohybrid, dominance, recessive, test cross, back cross, dihybrid, trihybrid and polyhybrid

15. Chromosomal theory of inheritance. Deviation from Mendelian inheritance - allelic interactions – dominance vs recessive, complete dominance, codominance, incomplete dominance,threshold characters.
16. Deviation from Mendelian inheritance – non allelic interaction without modification in Mendelian ratio (simple interaction) – Bateson and Punnett's experiment on fowl comb shape.
17. Non allelic interaction with modification in Mendelian ratio – i.) dominant epistasis (12:3:1)ii.) Recessive epistasis(9:3:4) iii.) duplicate and additive epistasis(9:6:1). iv.) duplicate dominant epistasis(15:1).
18. **Mid Semester Examination**
19. Duplicate recessive epistasis (9:7) vi.) dominant and recessive epistasis (13:3); summary of epistatic ratios.
20. Qualitative and quantitative traits - polygenes and continuous variation – transgressive segregation. Multiple factor hypothesis - Nilsson Ehle experiment on wheat kernel colour, corolla length in *Nicotiana longiflora*.
21. Lethal genes, pleiotropism, penetrance and expressivity, multiple alleles, blood group genetics in humans, coat colour in rabbits, self incompatibility in plants, pseudo alleles, isoalleles.
22. Crossing over mechanism– significance of crossing over, cytological proof for crossing over - Stern's experiment, factors controlling crossing over.
23. Linkage and its estimation - coupling and repulsion, experiment on Bateson and Punnet. Chromosomal theory of linkage of Morgan – complete and incomplete linkage, linkage group.
24. Strength of linkage and recombination, double cross over, interference and coincidence.
25. Two point and three point test cross genetic map, physical map.
26. Sex determination -autosomes and sex chromosomes, chromosomal theory of sex determination - sex determination in human, fowl, butterfly, grasshopper, honey bee, fumea. Sex determination in plants – *Melandrium*, papaya, maize. Genic balance theory of Bridges and gynandromorphs.
27. Sex linked inheritance – criss cross inheritance, reciprocal difference; holandric genes; sex influenced and sex limited traits.
28. Mutation - classification, methods of inducing mutations, mutagenic agents and induction of mutation.

29. Maternal effects – pattern of snail shell coiling .Cytoplasmic inheritance and– features of cytoplasmic inheritance.Chloroplast, mitochondrial - plastid colour in *Mirabilis jalapa* - cytoplasmic male sterility in maize.Infective particles - kappa particles in paramecium.
30. DNA, the genetic material – Griffith's experiment, experiment of Avery, McCleod and McCarty – confirmation by Hershey and Chase; RNA as genetic material – Frankel, Conrat and Singer experiment.
31. Types of DNA and RNA (mRNA, tRNA, rRNA) - structure of DNA – Watson and Crick model, proof for semi conservative
32. Models of DNA replication, steps involved in DNA replication.
33. Genetic code, cistron, muton and recon. Gene concept - Gene structure, molecular concept of gene, molecular definition of a gene, number of genes on a single chromosome, size of a gene, fine structure of a gene, types of genes.
34. Central dogma of life - Transcription and protein synthesis - regulation of gene expression – Operon model of Jacob and Monad – Lac and Trp operons.

Practical Schedule

1. Study of microscope
2. Study of cell structure.
3. Principles of killing and fixing; preparation of stains and preservatives. Study of behavior of chromosomes in mitosis -onion / *Aloe sp*
4. Procedure for fixing and observing different meiotic phases in the inflorescence of maize /rice and making temporary and permanent slides
5. Procedure for fixing and observing different meiotic phases in the inflorescence in sorghum /pearl millet and making temporary and permanent slides.
6. Chi square test and Probability analysis
7. Experiments on monohybrid (Dominance, Incomplete , codominance, dominant and recessive lethal) back cross and test cross.
8. Experiments on dihybrid, trihybrid, test cross and back cross
9. Experiments on simple interaction of genes-comb character in fowls; dominant epistasis.
10. Recessive epistasis, duplicate and additive epistasisincluding test cross and back cross.
11. Duplicate dominant epistasis, duplicate recessive epistasis, dominant and recessive epistasis including test cross and back cross.
12. Multiple alleles and polygenic inheritance
13. Determination of linkage with F_2 and test cross data; Coupling and repulsion.

14. Problems on two and three point test cross; Working out interference, coincidence and drawing genetic maps.
15. Problems on sex linked, limited and influenced inheritance in Drosophila and Human
16. Study on models on DNA and RNA structures.
17. **Final Practical examination.**

References

1. Fundamentals of Genetics: B. D. Singh , Kalyani Publishers
2. Genetics: M. W. Strickberger. New York : Macmillan
3. Principles of Genetics: Gardner, Simmons and Snustad.
4. Principles of Genetics: Sinnott, Dunn and Dobzhansk
5. Cytogenetics : Gupta P.K., Rastogi Publications, Meerut
6. Genetics : Verma, P.S. and V.K.Agarwal.
7. Theory and problems of genetics; Stansfield, W.D.
8. Elements of Genetics : Pundhan singh, Kalyani Publishers
9. Genes IX: Benjamin Lewin. Oxford University Press, Oxford.
10. Fundamentals of genetics: Russel, P.J. Addition Wesley Longman Publishers, USA
11. Introduction to Cytogenetics and Plant Breeding: Daniel Sundararaj, G. Thulasidas and M.Stephen Dorairaj, Popular Book Depot, Chennai –15.

AGR 201 Crop Production Technology-I (*Kharif* Crops) (1+2)

Theory

Unit I : Agro-Techniques for Rice & Maize

Origin, geographical distribution, economic importance, soil and climatic requirements, Varieties, cultural practices and yield of *Kharif* crops. Cereals- rice and maize

Unit II: Agro-Techniques for Millets & Pulses

Sorghum, pearl millet, finger millet and other minor millets, pulses- red garm, Green gram and black gram and cowpea

Unit III: Agro-Techniques for Oilseeds

Oilseeds: groundnut, soybean, sesame, castor

Unit IV - Agro-Techniques for Fibre Crops

Fiber crops and cotton and Jute

Unit V: Agro-Techniques for Forage Crops

Forage crops- sorghum, cluster bean, maize, guinea and cumbu Napier.

Practicals

Rice nursery preparation, transplanting of rice, sowing of soybean, pigeon pea and mungbean, maize, groundnut and cotton, effect of seed size on germination and seedling vigour of *Kharif* crops, effect of sowing depth on germination of *Kharif* crops, identification of weeds in *Kharif* crops, top dressing and foliar feeding of nutrients, study of yield contributing characters and yield calculation of *Kharif* crops, study of crop varieties and important agronomic experiments at experiential farm, recording biometric observations, Study of forage experiments, morphological description of *Kharif* crops, silage and hay making, visit to research centres of related crops.

Theory Schedule

1. Rice - Origin - Geographic distribution - Economic importance - Varieties - Soil and climatic requirement
2. Rice - Cultural practices - Yield - Economic benefits
3. Special type of rice cultivation - SRI and Hybrid rice cultivation
4. Maize - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield
5. Sorghum and Pearl millet - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield
6. Finger millet and Minor millets - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield

7. Pigeon pea - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield
8. Green gram, Black gram and Cowpea - Origin, geographic distribution, economic importance, soil and climatic requirement, varieties, cultural practices and yield - Agronomy of rice fallow pulses
9. **Mid-Semester Examination**
10. Groundnut - Origin, geographical distribution, economic importance, soil and climatic requirements - varieties, cultural practices yield and economics
11. Sesame and Soybean - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield
12. Castor- Origin, geographical distribution, economic importance, soil and climatic requirements - varieties, cultural practices yield and economics
13. Cotton - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield
14. Jute - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield
15. Fodder sorghum, Fodder cow pea - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield
16. Fodder maize, Cenchrus (Kollukattai pull):- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield
17. Cumbu Napier and, Guinea grass- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield study of rice ecosystems, climate, weather, seasons and varieties of Tamil Nadu

Practical Schedule

1. Practicing various nursery types, main field preparation and transplanting of rice crop
2. Nursery and main field preparation for important millets, pulses, oilseeds and fibre crops
3. Acquiring skill in different seed treatment techniques and its effect on seed size on germination and seedling vigour in important *kharif* crops
4. Acquiring skill in field preparation, sowing and manuring of crops under pure and intercropping situations for cereals and millets
5. Acquiring skill in field preparation, sowing and manuring of crops under pure and intercropping situations for pulses, oilseeds and forage crops
6. Identification of weeds in kharif crops
7. Acquiring skill in foliar nutrition for important kharif crops
8. Observations on growth parameters of cereals and millets
9. Observations on growth parameters of pulses, oilseeds & forage crops

10. Observations on yield parameters and estimation of yield in cereals and millets
11. Observations on yield parameters and estimation of yield in pulses and oilseeds
12. Observations on yield parameters and estimation of yield in forage crops
13. Study on silage and hay making for kharif crops
14. Visit to farmers field / research stations to study the cultivation techniques of cereal, millets, pulses, cotton, oilseeds and forage crops
15. Visit to important agronomic experiments and experimental farm for recording biometric and yield estimation
16. Working out cost and returns of important cereals, millets, pulses and oilseeds crops
17. Study of rice ecosystems, climate, weather, seasons and varieties of Tamil Nadu
18. Selection of nursery area, preparation of nursery, application of manures and fertilizer to nursery
19. Acquiring skill in seed treatment, seed soaking and incubation, nursery sowing and management
20. Study and Practice of main field preparation and puddling operations
21. Practicing of field preparatory operations - sectioning of field bunds and plastering, leveling and basal application of fertilizers
22. Practicing transplanting techniques in lowland rice
23. Estimation of plant population and acquiring skill in gap filling and thinning
24. Study of weeds and weed management in rice
25. Study and practice of green manuring and bio-fertilizer application in rice
26. Acquiring skill in nutrient management and practicing top dressing techniques
27. Study of water management practices for lowland rice
28. Observation of insect pests and diseases and their management
29. Recording growth and other related characters of rice
30. Estimation of yield and yield parameters in rice
31. Harvesting, threshing and cleaning and drying
32. Calculating the yield of produce
33. Working out cost of cultivation and economics
34. **Final Practical Examination**

References

1. Ahlawat, I.P.S., Om Prakash and G.S. Saini. 1998. Scientific Crop Production in India. Rama publishing House, Meerut.
2. Chidda Singh. 2010. Modern techniques of raising field crops. Oxford and IBH Publishing Co. Pvt. Ltd, New Delhi.

3. Singh. S.S. 2015. Crop management under irrigated and rainfed conditions. Kalyani Publishers, New Delhi.
4. Reddy, S.R. 2012. Agronomy of field crops. Kalyani publishers, New Delhi. Joshi, M. 2015. Textbook of Field crops. PHI Learning Pvt. Ltd., New Delhi ICAR 2015.
5. Hand Book of Agriculture. Indian Council of Agriculture, New Delhi Crop production Guide 2012. Directorate of Agriculture, Chennai.

e- Resources

1. www.crida.org.
2. www.cgiar.org.
3. www.tnau.ac.in/agriportal.

HOR 211 Production Technology for Fruit and Plantation Crops (1+1)

Theory

Unit I: Production status of fruit and plantation crops

Importance and scope of fruit and plantation crop industry in India – nutritional value of fruit crops - classification of fruit crops - area, production, productivity and export potential of fruit and plantation crops.

Unit II: Crop production techniques in tropical and sub tropical fruit crops

Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high - density planting - cropping systems - after care - training and pruning - water, nutrient and weed management –fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest.

Tropical Fruit crops : mango, banana, papaya, guava, sapota

Sub-tropical Fruit Crops : citrus (acid lime & sweet oranges), grape

Unit III: Crop production techniques in subtropical and Temperate fruit crops, Minor - fruits & nut Crops

Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high-density planting - cropping systems - after care - training and pruning - water, nutrient and weed management – fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest.

Sub-tropical Fruit Crops: Pineapple, pomegranate, jackfruit

Temperate Fruit crops: Apple, Pear, Peach, Plum Strawberry

Minor fruits & nut Crops: Ber, custard, amla, dragon fruit, nut crops (Almond, Pistachionut, walnut).

Climate and soil requirements – varieties – propagation and use of rootstocks - planting density and systems of planting -High density and ultra high-density planting -cropping systems - after care - training and pruning - water, nutrient and weed management –fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest.

Fruit crops: Apple, pear, peach, strawberry

Unit IV: Crop production techniques in palms and plantation crops

Importance and scope of plantation crop industry in India – area, production, productivity and export potential of plantation crops – commodity boards -Climate and soil requirements - varieties - propagation - nursery management - planting and – planting systems - cropping systems - after care - water, nutrient and weed management - intercropping - multi- tier cropping system - mulching - special horticultural practices - maturity indices, harvest and yield - pests and diseases - processing - value addition

Palms: Coconut, Arecanut, Oil palm, Palmyrah and Rubber

Unit V: Crop production techniques in beverage crops

Climate and soil requirements - varieties - propagation - nursery management - planting and – planting systems - cropping systems - after care - water, nutrient and weed management - intercropping - multi- tier cropping system - mulching - special horticultural practices - maturity indices, harvest and yield - pests and diseases - processing - value addition.

Beverage crops: Tea, Coffee, Cocoa, Cashew

Practicals

Propagation methods for fruit crops - description and identification of varieties - preparation of plant bio regulators & their uses – nutrient deficiency and disorders of fruit crops - fertilizers-application - pests and diseases- micro propagation in fruit crops- Visit to commercial orchard.

Fruit Crops: Mango, banana, papaya, guava, sapota, grapes, citrus (Mandarin and acid lime), pomegranate and jackfruit

Propagation methods for plantation crops - description and identification of plantation crops - preparation of plant bio regulators & their uses - nutritional disorders of plantation crops - fertilizers-application - pests and diseases- cost economics of plantation crops. Visit to plantations and plantation industries.

Palms and plantation Crops: Coconut, Arecanut, Cashew, Tea, Coffee, Rubber and Cocoa

Theory schedule

1. Importance and scope of fruit and plantation crop industry in India nutritional value of fruit crops - Classification of fruit crop, area, production, productivity and export potential of fruit and plantation crops
2. Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high - density planting - cropping systems - after care - training and pruning - water, nutrient and weed management – fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest of Mango
3. Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high - density planting - cropping systems - after care - training and pruning - water, nutrient and weed management –

fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest of Banana

4. Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high - density planting - cropping systems - after care - training and pruning - water, nutrient and weed management – fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest of Papaya, Guava and sapota
5. Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high - density planting - cropping systems - after care - training and pruning - water, nutrient and weed management – fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest of Grapes
6. Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high - density planting - cropping systems - after care - training and pruning - water, nutrient and weed management – fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest of Citrus (Sweet orange and Acid Lime), and pineapple
7. Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high - density planting - cropping systems - after care - training and pruning - water, nutrient and weed management – fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest of Jackfruit and Pomegranate
8. Climate and soil requirements – varieties – propagation and use of rootstocks- planting density and systems of planting - High density and ultra high - density planting - cropping systems - after care - training and pruning - water, nutrient and weed management – fertigation - special horticultural techniques - plant growth regulation - important disorders – maturity indices and harvest of Minor fruits Crop – Ber and Custard apple, amla, dragon fruit and nut crops (Almond, Pistachionut, walnut),
9. **Mid semester examination**
10. Climate and soil – varieties - propagation methods - planting and cropping systems - after care-training and pruning - water, nutrient and weed management - plant growth regulation - important disorders – maturity indices and harvest- post harvest management of Apple and pear Peach and strawberry

11. Importance and scope of plantation crop industry in India – area, production, productivity and export potential of plantation crops – commodity boards.
12. Climate and soil requirements - varieties - propagation - nursery management - planting systems - planting density -nutrient, water and weed management - intercropping at various ages of plantation -multitier cropping - shade management - nutritional disorders - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition of Coconut
13. Climate and soil requirements - varieties - propagation - nursery management - planting systems - planting density - nutrient, water and weed management - intercropping at various ages of plantation - multitier cropping – shade management - nutritional disorders - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition of Arecanut and Cocoa .
14. Climate and soil requirements - varieties – propagation - nursery management - planting - nutrient, water and weed management - water conservation techniques - leaf pruning - pollination - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition Oil palm and Palmyrah.
15. Climate and soil requirements - varieties - propagation - nursery management - planting and planting density - HDP - UHDP - nutrient, water and weed management - cover cropping - tapping - use of plant growth regulators - top working - maturity indices - harvest and yield , latex yield and processing - pests and diseases - grading - processing and value addition Rubber and Cashew
16. Climate and soil requirements- varieties – propagation - nursery management - planting density and systems of planting - nutrient, water and weed management - mulching - cropping systems - shade regulation - training and pruning - role of growth regulators - nutritional disorders - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition of Tea .
17. Climate and soil requirements - varieties – propagation - nursery management - planting - nutrient, water and weed management - mixed and inter cropping - shade management - training and pruning - role of growth regulators - nutritional disorders - maturity indices - harvest and yield - pests and diseases - grading - processing and value addition of Coffee.

Practical schedule

1. Propagation techniques, selection of planting material, varieties, important cultural practices for Mango

2. Propagation techniques, selection of planting material, varieties, important cultural practices for Banana
3. Propagation techniques, selection of planting material, varieties, important cultural practices for Papaya
4. Propagation techniques, selection of planting material, varieties, important cultural practices for Guava
5. Propagation techniques, selection of planting material, varieties, important cultural practices for Sapota
6. Propagation techniques, selection of planting material, varieties, important cultural practices for Grapes
7. Propagation techniques, selection of planting material, varieties, important cultural practices for Citrus (Sweet orange and acid lime)
8. Propagation techniques, selection of planting material, varieties, important cultural practices for Pomegranate
9. Propagation techniques, selection of planting material, varieties, important cultural practices for Jackfruit
10. Preparation and application of PGR's for propagation and fruit production
11. Nutrient deficiencies and its corrective measures in fruit crops
12. Identification and description of varieties - mother palm and seed nut selection – nursery practices seedling selection -fertilizers application – nutritional disorders of coconut
13. Identification and description of varieties - mother palm and seed nut selection – nursery practices seedling selection -fertilizers application – nutritional disorders of Arecanut and Cocoa
14. Identification and description of varieties - nursery practices - training and pruning - processing of Tea and coffee
15. Identification and description of varieties, clones - bud wood nursery practices - propagation techniques - top working – processing of Rubber and cashew
16. Visit to commercial orchard and plantation industries.
17. **Practical examination**

References

1. Banday, F.A. and Sharma, M.K.2010 Advances in temperate fruit production. Kalyani Publishers, Ludhiana

2. Bose, T.K., S.K. Mitra and D. Sanyal 2001. Fruits: Tropical and Subtropical (2 volumes) Naya Udyog, Calcutta.
3. Bose, T.K., S.K. Mitra, A.A. Farooqi and M.K. Sadhu (Eds). 1999. Tropical Horticulture Vol.1. Naya Prokash, Calcutta.
4. Chadha, K.L. 2001. Handbook of Horticulture. ICAR, Delhi
5. Chadha, T.R. 2001 Textbook of temperate fruits. ICAR, New Delhi
6. Chattopadhyay, T.K. 2001. A Text Book on Pomology (4 volumes). Kalyani Publishers, Ludhiana.
7. Chattopadhyay. 1998. A textbook on pomology (sub-tropical fruits) vol.III. Published by M/s. Kalyani publishers, Ludhiana, New Delhi, Noida. UP.
8. Chudawat, B. S.1990. Arid fruit culture Oxford &IBH, New Delhi
9. Das, B.C. and Das S.N. Cultivation of minor fruits. Kalyani Publishers, Ludhiana
10. David Jackson and N.E. Laone, 1999. Subtropical and temperate fruit production. CABI
11. Publications
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AEX 201 Fundamentals of Extension Education (1+1)

Theory

Unit I: Extension education and programme planning

Education: Meaning, definition and Types; Extension Education: meaning, definition, scope and process; objectives and principles of Extension Education; Extension Programme planning: Meaning, Process, Principles and Steps in Programme Development

Unit II: Extension systems in India

Extension systems in India: extension efforts in pre-independence era (Sriniketan, Marthandam, Firka Development Scheme, Gurgaon Experiment, etc.) and post-independence era (Etawah Pilot Project, Nilokheri Experiment, etc.); Reorganised Extension System (T&V system) various extension/ agriculture development programs launched by ICAR/ Govt. of India (IADP, IAAP, HYVP, KVK, IVLP, ORP, ND, NATP, NAIP, etc.). Social Justice and poverty alleviation programme: ITDA, IRDP/SGSY/NRLM. Women Development Programme: RMK, MSY etc.

Unit III: New trends in agriculture extension& Rural Development

New trends in agriculture extension: privatization extension, cyber extension/ e-extension, market-led extension, farmer-led extension, expert systems, etc., DWCRA, Commodity Interest Groups (CIGs)., Farmers Producer Group (FPG).

Unit IV: Rural Development& Community Development

Rural Development: concept, meaning, definition; various rural development programs launched by Govt. of India. Community Development: meaning, definition, concept and principles, Philosophy of C.D. Rural Leadership: concept and definition, types of leaders in rural context; Method of identification of Rural Leader. Extension administration: meaning and concept, principles and functions. Monitoring and evaluation: concept and definition, monitoring and evaluation of extension programs; transfer of technology: concept and models, capacity building of extension personnel.

Unit V: Extension teaching methods& Agricultural Journalism

Extension Teaching Methods: Meaning, classification, individual, group and mass contact methods, ICT Applications in TOT (New and social media), media mix strategies; Principles and Functions of Communication,Agriculture journalism; diffusion and adoption of innovation: concept and meaning,attributes of innovation, process and stages of adoption, adopter categories.

Practicals

To get acquainted with university extension system, Group discussion- exercise, Identification of rural leaders in village situation; preparation and use of AV aids, digital camera and

Smartboard, preparation of extension literature-leaflet, booklet, folder, pamphlet, newstories and success stories, Presentation skills exercise: micro teaching exercise: A visit to village to understand the problems being encountered by the villagers/ farmers : to study organization and functioning of DRDA/PRI and other development departments at district level: visit to NGO/FO/FPO and learning from their experience in rural development: understanding PRA techniques and their application in village development planning: exposure to mass media; visit to community radio and television studio for understanding the process of programme production: Script writing, writing for print and electronic media, developing script for radio and television.

Theory Schedule

1. Education: Meaning, definition and Types; Extension Education: meaning, definition, scope and process; objectives and principles of Extension Education
2. Extension Programme planning – Meaning, definition, process, principles and steps in programme planning / development.
3. Extension systems in India: Extension efforts in pre-independence era (Sriniketan, Marthandam, Firka Development Scheme, Gurgaon Experiment, etc.) and post-independence era (Etawah Pilot Project, Nilokheri Experiment, etc)-Reorganised Extension System (T&V system).
4. Various extension/ agriculture development programs launched by ICAR/ Govt. of India (IADP, IAAP, HYVP, KVK, IVLP, ORP, ND, NATP, NAIP, etc.). Social Justice and poverty alleviation programme: ITDA, IRDP/SGSY/NRLM) & Women Development Programme: RMK, MSY etc.
5. New trends in agriculture extension: privatization of extension, cyber extension/ e-extension
   (Video conferencing, Podcast, e-Velanmai, M-Velanmai app. and multimedia).
6. Market-led extension, farmer-led extension: Meaning, definition and importance in extension; expert systems: meaning, definition and application in agriculture (Expert system apps).
7. Rural Development: concept, meaning, definition; various rural development programs launched by Govt. of India (DWCRA, MGNREGA,RSVY, IAY, SGRY, ITDP and PMGSY).
8. **Mid Semester Examination**
9. Community Development: meaning, definition, concept and principles, Philosophy of Community development.
10. Rural leadership: concept and definition, types of leaders in rural context; Methods of identification of Rural Leader.

11. Extension administration: meaning and concept, principles and functions. Monitoring and evaluation: concept and definition, monitoring and evaluation of extension programs & types of evaluation.
12. Transfer of technology: concept and models (Top down Model, Feedback Model, Farmer- back to - Farmer Model and REP Interaction model), capacity building of extension personnel: Training, meaning, definition and types of trainings
13. Extension teaching methods: meaning, classification; Individual methods- Farm and Home, Personal letter, Official call, observation and Result demonstration.
14. Group contact methods - Method demonstration, meeting, lecture, debate, workshop, seminar, forum, conference, symposium, panel, brain storming, buzz session, role playing and simulation games.
15. Mass contact methods- Campaign, exhibition, farmers day and field trip; ICT Applications in TOT (New and social media), media mix strategies;
16. Principles and Functions of Communication -Agricultural Journalism(Print media) - definition, principles, importance, ABC of news, types of news.
17. Diffusion and Adoption of innovation – concept and meaning, attributes of innovation, process and stages of adoption, adopter categories.

Practical Schedule

1. Study the University Extension system
2. Exercise on practicing group discussion technique
3. Identification of rural leaders in village situation
4. Handling and use of AV aids, digital camera and smartboard
5. Preparation of extension literature- leaflet, folder, booklet and pamphlet
6. Preparation of news stories and success stories
7. Exercise on Presentation skill and microteaching
8. Visit a village and understand the socio cultural and agricultural related problems being encountered by the villagers/ farmers
9. Study the organizational set up and functions of DRDA/PRI
10. Visit to State department of Agri/ Horti to understand the organizational setup, roles, functions and various schemes
11. Visit to NGO/FO/FPO and learning from their experience in rural development
12. Understanding PRA techniques and their application in village development planning
13. Visit to Community Radio station/ Educational Media Centre to understand the process

of programme production

14. Visit to All India Radio Station / TV studio/News Agency/ Press to study the various activities & programmes.
15. Exercise on Script writing and developing an extension programme for Radio
16. Exercise on Script writing and developing an extension programme for TV/Social Media
17. **Final Practical Examination**

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NEM 201 Fundamentals of Nematology (1+1)

Theory

Unit I: History and Development of Nematology and Importance of Nematodes

Introduction – History of Phytonematology, habitat and diversity, economic importance of nematodes.

Unit II: Morphology and Taxonomy of Nematodes

General characteristics of plant parasitic nematodes. Nematode definition, general morphology and biology Classification of nematodes up to family level with emphasis on groups containing economically important genera. Classification of nematodes on the basis of feeding/parasitic habit.

Unit III: Symptoms, Interaction and Bio-ecology of nematodes

Symptomatology, role of nematodes in disease development, Interaction between plant parasitic nematodes and disease causing fungi, bacteria and viruses.

Unit IV: Nematode pests of crops

Nematode pests of crops: Rice, wheat, vegetables, pulses, oilseed and fiber crops, citrus and banana, tea, coffee and coconut

Unit V: Nematode management

Different methods of nematode management: Cultural methods, physical, methods, Biological methods, Chemical methods, Plant Quarantine, Plant resistance and INM.

Practicals

Sampling methods, collection of soil and plant samples; Extraction of nematodes from soil and plant tissues following Cobb's sieving and decanting technique, Baermann's funnel technique, Picking and counting of plant parasitic nematode. Identification of economically important plant nematodes up to generic level with the help of keys and description: *Meloidogyne*, *Pratylenchus*, *Heterodera*, *Tylenchulus*, *Xiphinema*, and *Helicotylenchus* etc. Study of symptoms caused by important nematode pests of cereals, vegetables, pulses and plantation crops etc. Methods of application of nematicides and organic amendments.

Theory Schedule

1. Introduction – Brief history and development of nematology at National and International levels.
2. Position of nematodes in animal kingdom – Importance of nematodes (human being, animals and plants, Parasites of insects) – habitat and diversity - Economic loss in crop plants.

3. General morphology of nematodes.
4. Anatomy of nematodes – Digestive, excretory and nervous system and sense organs.
5. Anatomy of nematodes – Reproductive system.
6. Biology of economically important nematode genera *Meloidogyne*, *Heterodera*, *Tylenchulus*, *Rotylenchulus* and *Radopholus.*
7. Taxonomy of plant – parasitic nematodes of the Secernentea and Adenophorea.
8. Classification of plant – parasitic nematodes based on feeding habits – Beneficial nematodes.
9. **Mid Semester Examination**
10. Symptoms of nematode damage and role of nematodes in disease development.
11. Interaction between plant parasitic nematodes and disease causing fungi, bacteria and viruses.
12. Nematode parasites of cereals.
13. Nematode parasites of vegetables crops
14. Nematode parasites of pulses, oilseed and fiber crops.
15. Nematode parasites of fruit and plantation crops.
16. Principles of nematode management (Legislative, physical, cultural, biological, chemical and host-plant resistance).
17. Integrated nematode management.

Practical Schedule

1. Soil and root sampling. Extraction of nematodes by Cobb's sieving method; Baermann funnel Technique and modified Baermann funnel technique.
2. Extraction of nematodes by sugar flotation technique; Extraction of cysts by conical flask technique and fenwick can method.
3. Extraction of nematodes from roots and staining of roots infested with endoparasitic and semi – endoparasitic nematodes.
4. Preservation of nematodes and preparation of temporary and permanent slides.
5. Observing morphology of the order Tylenchida (*Hoplolaimus*) and Dorylaimida (*Xiphinema*, *Longidorus*).
6. Identification of nematodes – *Tylenchorhynchus, Helicotylenchus*. *Hoplolaimus*
7. Identification of nematodes – *Pratylenchus*, *Radopholus* and *Hirschmanniella*.
8. Identification of nematodes – *Hemicriconemoides* – *Criconema*, *Heterodera* – *Globodera*.

9. Identification of nematodes – *Tylenchulus* and *Rotylenchulus*
10. Study of life stages of sedentary endoparasitic nematodes
11. Study of life stages of semi and migratory endoparasitic nematodes.
12. Nematode damage symptoms of Cereals, Pulses, Oilseeds and Fiber crops
13. Nematode damage symptoms of Vegetables crops.
14. Nematode damage symptoms of Fruit crops.
15. Nematode damage symptoms of Plantation crops.
16. Study of types of nematicides, application methods and calculation of dosages; study of biocontrol agents.
17. **Final Practical examination.**

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AGR 202 Principles and Practices of Natural Farming (1+1)

Theory

Unit I: Indian Heritage of Ancient Agriculture and the Evolution of Natural Farming: An Ecological and Socio-Cultural Perspective

Indian Heritage of Ancient Agriculture: Study of ancient Indian agricultural practices, indigenous wisdom, and traditional methods of soil, water, and crop management that form the roots of sustainable farming systems. History and Evolution of Natural Farming- Traces the global and Indian journey of natural farming — from traditional practices to modern systems like Zero Budget Natural Farming and Fukuoka's "Do-nothing" farming philosophy. Concept, Definition, and Objectives of Natural Farming - Explains what natural farming means, its ecological and socio-economic objectives, and its importance in addressing climate change, food security, and rural sustainability.

Unit II: Natural Farming Paradigms: Core Principles, Foundational Pillars, and Comparative Evaluation of Global and Indian Models

Essential Characteristics and Principles of Natural Farming - the key principles of natural farming — no tillage, no chemical inputs, minimal disturbance, and reliance on natural processes for fertility and pest control. Main Pillars of Natural Farming - Discusses the four main pillars (Bijamrit, Jivamrit, Mulching, and Waaphasa) and their scientific basis, along with other supporting practices ensuring soil and plant health. Scope and Importance of Natural Farming - Highlights the economic, environmental, and social significance of natural farming in modern agriculture and its contribution to achieving sustainable development goals. Methods and Schools of Natural Farming - Introduces various natural farming approaches such as Fukuoka Natural Farming, Subhash Palekar's ZBNF, Rishi Krishi, and Korean Natural Farming, emphasizing their unique principles.

Unit III: Design Principles of Natural Farms: Ecological Balance, Integrated Systems, and Biodiversity Conservation, recycling of farm waste

Characteristics and Design of a Natural Farm- Describes how a natural farm is structured — including crop diversity, livestock integration, and natural resource use — to maintain ecological balance and productivity. Ecological Balance, Ecological Engineering, and Community Responsibility- explores how natural farming systems maintain ecological harmony and the collective role of communities in promoting sustainability and biodiversity conservation. Integration of Crops, Trees, and Animals in Natural Farming - Focuses on integrated farming systems that combine crops, livestock, and trees for better resource recycling, resilience, and farm sustainability. Biodiversity, Indigenous Seeds, and Farm Waste Recycling - Emphasizes the importance of conserving native species, using indigenous seeds, and converting farm wastes into valuable inputs for soil enrichment.

Unit IV: Resource-Efficient Natural Farming: Water Conservation, Renewable Energy Integration, Nutrient Cycling, and Ecological Pest Management

Water Conservation, Renewable Energy, and Nutrient Management in Natural Farming - Discusses efficient water use, renewable energy applications, and natural nutrient sources such as composts, Jivamrit, and crop residues. Insect, Pest, Disease, and Weed Management in Natural Farming - Describes ecological pest management techniques, use of botanical formulations, and biodiversity-based prevention methods without synthetic pesticides.Mechanization and Farm Operations in Natural Farming - Covers suitable tools, machines, and low-cost technologies adapted to natural farming while maintaining minimal soil and ecosystem disturbance.

Unit V: Value Chain Development in Natural Farming: Processing, Labelling, Certification Standards, and Market Opportunities

Processing, Labelling, and Economic Viability of Natural Farming - Examines post-harvest handling, value addition, packaging, and labeling practices that enhance market value and ensure farmer profitability. Certification, Standards, and Export Potential of Natural Farming Produce- Details certification systems like PGS and third-party standards, quality assurance mechanisms, and export opportunities for natural farm products.Government Initiatives, Case Studies, and Entrepreneurship Opportunities in Natural Farming - Reviews national and state-level programs, NGO initiatives, successful farmer case studies, and emerging entrepreneurial avenues in natural and chemical-free agriculture.

Practicals

Visit of natural farm and chemical free traditional farms to study the various components and operations of natural farming principles at the farm; Indigenous technical knowledge (ITK) for seed, tillage, water, nutrient, insect-pest, disease and weed management; On-farm inputs preparation methods and protocols, Studies in green manuring in-situ and green leaf manuring, Studies on different types of botanicals and animal urine and dung based non-aerated and aerated inputs for plant growth, nutrient, insect and pest and disease management; Weed management practices in natural farming; Techniques of Indigenous seed production- storage and marketing, Partial and complete nutrient and financial budgeting in natural farming; farming; Evaluation of ecosystem services in natural farming (Crop, Field and System).

Theory schedule

1. Indian Heritage of Ancient Agriculture: Study of ancient Indian agricultural practices, indigenous wisdom, and traditional methods of soil, water, and crop management that form the roots of sustainable farming systems.

2. History and Evolution of Natural Farming- Traces the global and Indian journey of natural farming — from traditional practices to modern systems like Zero Budget Natural Farming and Fukuoka's "Do-nothing" farming philosophy.
3. Concept, Definition, and Objectives of Natural Farming - Explains what natural farming means, its ecological and socio-economic objectives, and its importance in addressing climate change, food security, and rural sustainability.
4. Essential Characteristics and Principles of Natural Farming - the key principles of natural farming — no tillage, no chemical inputs, minimal disturbance, and reliance on natural processes for fertility and pest control.
5. Main Pillars of Natural Farming - Discusses the four main pillars (Bijamrit, Jivamrit, Mulching, and Waaphasa) and their scientific basis, along with other supporting practices ensuring soil and plant health.
6. Scope and Importance of Natural Farming - Highlights the economic, environmental, and social significance of natural farming in modern agriculture and its contribution to achieving sustainable development goals.
7. Methods and Schools of Natural Farming - Introduces various natural farming approaches such as Fukuoka Natural Farming, Subhash Palekar's ZBNF, Rishi Krishi, and Korean Natural Farming, emphasizing their unique principles.
8. Characteristics and Design of a Natural Farm- Describes how a natural farm is structured — including crop diversity, livestock integration, and natural resource use — to maintain ecological balance and productivity.
9. **Mid Semester Examination**
10. Ecological Balance, Ecological Engineering, and Community Responsibility- explores how natural farming systems maintain ecological harmony and the collective role of communities in promoting sustainability and biodiversity conservation.
11. Integration of Crops, Trees, and Animals in Natural Farming - Focuses on integrated farming systems that combine crops, livestock, and trees for better resource recycling, resilience, and farm sustainability.
12. Biodiversity, Indigenous Seeds, and Farm Waste Recycling - Emphasizes the importance of conserving native species, using indigenous seeds, and converting farm wastes into valuable inputs for soil enrichment.
13. Water Conservation, Renewable Energy, and Nutrient Management in Natural Farming - Discusses efficient water use, renewable energy applications, and natural nutrient sources such as composts, Jivamrit, and crop residues.

14. Insect, Pest, Disease, and Weed Management in Natural Farming - Describes ecological pest management techniques, use of botanical formulations, and biodiversity-based prevention methods without synthetic pesticides.
15. Mechanization and Farm Operations in Natural Farming - Covers suitable tools, machines, and low-cost technologies adapted to natural farming while maintaining minimal soil and ecosystem disturbance.
16. Processing, Labelling, and Economic Viability of Natural Farming - Examines post-harvest handling, value addition, packaging, and labeling practices that enhance market value and ensure farmer profitability.
17. Certification, Standards, and Export Potential of Natural Farming Produce- Details of certification systems like PGS and third-party standards, quality assurance mechanisms, and export opportunities for natural farm products. Government and NGO Initiatives, Case Studies, and Entrepreneurship Opportunities in Natural Farming.

Practical Schedule

1. Visit of natural farming field at campus/ farmers holdings and study the various on -farm and off farm resources
2. Study the various components and operations of natural farming principles at the farm level
3. Studies on chemical free traditional farms and preparation of botanical extract for foliar applications for crop growth and development
4. Acquiring skill on Indigenous technical knowledge (ITK) for seed treatments, practicing of sowing in nursery and observed the germinability of the treated seeds
5. Studies on Indigenous technical knowledge (ITK) for tillage, water and nutrient management
6. Acquiring skills on insect-pest, disease and weed management by using ITK’s
7. Acquiring skills pertaining to On-farm inputs preparation methods and also protocols for preparation of composting techniques
8. Acquiring skills on vermicompost preparation and their effect on crop production
9. Studies in green manuring in-situ incorporation and practicing green leaf manuring application in the field
10. Preparation of Non-Aerated Bio-Inputs (Beejamrit, Jivamrit, and Cow Urine Extracts) and learning the foliar application of these bio inputs
11. Preparation and Application of Aerated Compost Teas and Fermented Botanical Extracts
12. Studies on weed management practices in natural farming ecosystems
13. Studies on the techniques of Indigenous seed production- storage and marketing strategies

14. work out the Partial and complete nutrient and financial budgeting in natural farming and conventional farming situation
15. work out the cost of cultivation for organic rice/pulse cultivation
16. visit to natural farming/ organic farming field for studying of ecosystem services in natural farming (Crop, Field and System).
17. **Final Practical Examination**

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SAC 201 Soil Resource Inventory and Geoinformatics (1+0)

Theory

Unit-I: Soil Resource Inventory

Definition- principles, concepts - Scope and objectives - Fundamental and Applied. Soil systematics- pedon and polypedon, control section and soil correlation; Soil mapping units: Soil series, soil association, soil complex, variants, inclusions and miscellaneous land types.

Unit II: Soil survey concepts

standard soil survey, Methods of soil survey: Base maps, Traversing: Grid survey and Free survey. Types of soil survey: Detailed, Reconnaissance, Detailed- Reconnaissance and Semi–Detailed soil survey.

Unit-III: Soil survey reports

Soil Survey Interpretations –Land Capability Classification –Soil and Land Irrigability Classification- Storie Index Rating –Productivity potential- Fertility Capability Classification-Crop suitability: Field crops, horticultural crops and forest trees. Delineation of soils for fertility – Nutrient indexing. Land Use Planning concepts and objectives.

Unit-IV: Remote sensing concepts & it's types

Electromagnetic radiation (EMR): principles, EMR interaction with atmosphere and earth surface features- Platforms and sensors. Introduction to basemaps & satellite data products (cadastral map, toposheet, aerial photograph & satellite images) & image interpretation elements.

Unit-V: Geographic Information System (GIS)

Definition, components and functions; Global Positioning System (GPS / GNSS) – functions, segments and working principles, GPS errors. Geoinformatics applications in agriculture - Land use land cover mapping - Soil resource mapping.

Theory Schedule

1. Soil resource Inventory -Definition- principles, concepts; Soil systematics - pedon and polypedon, control section and soil correlation
2. Standard soil survey - scope and objectives; Fundamental and applied.
3. Soil mapping units: Soil series, soil association, soil complex, variants, inclusions and miscellaneous land types
4. Soil survey concepts - Methods of soil survey: Base maps, Traversing: Grid survey and Free survey
5. Types of soil survey: Detailed, Reconnaissance, Detailed- Reconnaissance and Semi–Detailed soil survey
6. Interpretative groupings of Soil: Land Capability Classification and Fertility Capability Classification.

7. Interpretative groupings of Soil: Land Irrigability Classification, Storie index and Productivity potential.
8. Land Suitability Classification for field crops, horticultural crops and forest trees
9. **Mid Semester Examination**
10. Land Use Planning: Concepts and objectives; land use planning for Tropical, subtropical and temperate regions
11. Remote Sensing - Definition and Components; Remote sensing platforms and sensors
12. Electromagnetic Radiation (EMR) – Principle, Interaction of EMR in the atmosphere
13. Interaction of EMR with earth surface features (Soil, Water and Vegetation)
14. Introduction to basemaps & satellite data products (cadastral map, toposheet, aerial photograph & satellite images) & image interpretation elements
15. GIS - definition and components; Raster and vector data models
16. GPS and GNSS – functions, segments, working principle and GPS errors
17. Geoinformatics in Agricultural applications (soil & crop studies)

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3. www.iuss.org/Bulletins/00000096.pdf
4. www.oosa.unvienna.org/pdf/sap/centres/rscurrE.pdf -

AEC 201 Farm Management, Production and Resource Economics (1+1)

Theory

Unit I: Production Economics and Farm Management - Nature and Scope

Farm management: meaning, definitions and concepts; Nature and scope, Production economics and farm management principles; Meaning definition of production economics, objectives and relationship with other sciences; Basic concepts in production economics and farm management. Farm Decisions making process; Meaning and definition of farms sizes based on holding and ownership; Types of farming: Specialized farming, Mixed farming, Diversified farming and their characteristics, Systems of farming: Peasant Farming, State Farming, Capitalistic farming, Collective and Co – operative Farming, factors determining types and size of farms.

Unit II: Factor – Product, Factor – Factor and Product – Product Relationships

Meaning and definition of production function and its types, use of production function in decision making on a farm, factor-product relationship Meaning, Definition – Laws of Returns. Cost principle: meaning and concept of costs, types of costs-seven costs and applied cost concepts, and their interrelationship, importance of cost in managing farm business , shut down and break-even points, estimation of gross farm income, net farm income, family labor income and farm business income. Economies of Scale – Economies of Size - Determination of Optimum Input and Output – Physical and Economic Optimum. Factor-factor relationship: Least cost combination of inputs. Product-product relationship: Revenue maximizing combination of outputs. Law of equi-marginal returns or principles of opportunity cost and Minimum Loss Principle and law of comparative advantage.

Unit III: Farm Planning and Budgeting

Farm business analysis: meaning and concept of farm income and profitability, technical and economic efficiency measures in crop and livestock enterprises. Farm records: types and importance of farm records and accounts in managing a farm; various types of farm records needed to maintain on farm, farm inventory, balance sheet and profit and loss statement. Farm planning and budgeting: meaning and importance of farm planning and budgeting, partial and complete budgeting, steps in farm planning and budgeting, linear programming, appraisal of farm resources, selection of crops and livestock enterprises.

Unit IV: Risk and Uncertainty in Agriculture Production

Risk and uncertainty: concept of risk and uncertainty in agriculture production, types/sources of risks and their management strategies. Crop/livestock/machinery insurance: Weather based crop insurance (WBCIS) and Pradan Mantri Fasal Bhima Yojana (PMFBY), their features.

Unit V: Resource Economics

Resource economics: Meaning of resource economics, difference between Natural Resource Economics (NRE) and agricultural economics, natural resources: Definition, classification and unique properties of natural resources. Property Rights: Common Property Resources (CPRs): meaning and characteristics of CPRs, positive and negative externalities in agriculture, inefficiency and welfare loss, solutions, management of common property resources of land, water, pasture, fishery and forest resource.

Practicals

Study and visit to different farm layouts and appraisals of farm resources; Analysis of costs and revenue concepts; Computation of depreciation cost of farm assets; Classical Production Function and demarcation of three regions, Determination of most profitable level of input use in a farm production process; Determination of least cost combination of inputs; Selection of most profitable enterprise combination; Application of equi-marginal returns/opportunity cost principle in allocation of farm resources; Application of the principle of comparative advantage; Estimation of cost and returns using CACP cost concepts for crop, horticulture and livestock enterprises; Break Even Analysis, Farm inventory analysis; Preparation of optimum farm plan using budgeting technique using partial and complete budgeting; visit to farms to study farm records and accounts; Graphical solution to Linear Programming problem, Collection and analysis of data on various resources in India; Practical Examination.

Theory Schedule

1. Farm management: meaning, definitions and concepts; Nature and scope, Production economics and farm management principles; Meaning definition of production economics, objectives and relationship with other sciences
2. Basic concepts in production economics and farm management. Farm Decisions making process; Meaning and definition of farms sizes based on holding and ownership
3. Types of farming: Specialized farming, Mixed farming, Diversified farming and their characteristics; Systems of farming: Peasant Farming, State Farming, Capitalistic farming, Collective and Co – operative Farming. factors determining types and size of farms
4. Meaning and definition of production function and its types, use of production function in decision making on a farm, factor-product relationship Meaning, Definition – Laws of Returns
5. Cost principle: meaning and concept of costs, types of costs-seven costs and applied cost concepts, and their interrelationship, importance of cost in managing farm business,

shut down and break-even points, estimation of gross farm income, net farm income, family labor income and farm business income

6. Economies of Scale – Economies of Size - Determination of Optimum Input and Output – Physical and Economic Optimum. Factor-factor relationship: Least cost combination of inputs
7. Product-product relationship: Revenue maximizing combination of outputs. Law of equi-marginal returns or principles of opportunity cost and Minimum Loss Principle and law of comparative advantage
8. Farm business analysis: meaning and concept of farm income and profitability, technical and economic efficiency measures in crop and livestock enterprises. Farm records: types and importance of farm records and accounts in managing a farm
9. **Mid Semester Examination**
10. Various types of farm records needed to maintain on farm, farm inventory, balance sheet and profit and loss statement
11. Farm planning and budgeting: meaning and importance of farm planning and budgeting, partial and complete budgeting, steps in farm planning and budgeting, linear programming
12. Appraisal of farm resources, selection of crops and livestock enterprises
13. Risk and uncertainty: concept of risk and uncertainty in agriculture production, types/sources of risks and their management strategies
14. Crop/livestock/machinery insurance: Weather based crop insurance (WBCIS) and Pradan Mantri Fasal Bhima Yojana (PMFBY), their features
15. Resource economics: Meaning of resource economics, difference between Natural Resource Economics (NRE) and agricultural economics, Natural resources: Definition classification and unique properties of natural resources
16. Property Rights: Common Property Resources (CPRs): meaning and characteristics of CPRs
17. Positive and negative externalities in agriculture, inefficiency and welfare loss, solutions, management of common property resources of land, water, pasture, fishery and forest resource

Practical Schedule

1. Study and visit to different farm layouts and appraisals of farm resources
2. Computation of depreciation cost of farm assets
3. Classical Production Function and demarcation of three regions
4. Determination of most profitable level of input use in a farm production process
5. Determination of least cost combination of inputs
6. Selection of most profitable enterprise combination
7. Application of equi-marginal returns/opportunity cost principle in allocation of farm resources; Application of the principle of comparative advantage
8. Analysis of costs and revenue concepts: Estimation of cost and returns using CACP cost concepts for Agricultural crops
9. Estimation of cost and returns for horticultural crops
10. Estimation of cost and returns for livestock enterprises
11. Break Even Analysis
12. Farm inventory analysis: Preparation of optimum farm plan using partial budgeting
13. Farm inventory analysis: Preparation of optimum farm plan using complete budgeting
14. Visit to farms to study farm records and accounts
15. Graphical solution to Linear Programming problem
16. Collection and analysis of data on various resources in India; Practical Examination
17. **Final Practical Examination**

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6. Raju, V.T. and Vishwashankar Rao, Economics of Farm Production and Management.
7. Sadhu and Singh, Fundamentals of Agricultural Economics.
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3. https://agrigyan.in/farm-management-production-resource-economics-pdf-download/
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SWE 211 Fundamentals Soil and Water Conservation Engineering (1+1)

Theory

Unit I: Surveying

Surveying and levelling – chain, Cross staff survey – levelling – land measurement and computation of area – Simpson's rule and Trapezoidal rule.

Unit II: Soil erosion

Soil Erosion – causes and evil effects of soil erosion – geologic and accelerated erosion - water erosion -stages of water erosion - splash, sheet, rill and gully erosion - ravines - Soil loss estimation by universal soil loss equation - erosivity and erodibility – land slides – wind erosion - factors influencing wind erosion - mechanics of wind erosion – suspension, saltation, surface creep

Unit III: Soil conservation and Water harvesting

Erosion control measures for agricultural lands – biological measures – contour cultivation – strip cropping – cropping systems – vegetative barriers - windbreaks and shelterbelts - shifting cultivation - mechanical measures – contour bund – graded bund – broad beds and furrows – basin listing – random tie ridging – mechanical measures for hill slopes – contour trench – bench terrace – contour stone wall – Water harvesting techniques -Rain water harvesting – runoff water harvesting - Runoff Computation - Groundwater Recharge techniques -– Farm ponds and percolation ponds

Unit IV: Irrigation Water measurement and drainage

Irrigation - measurement of flow in open channels - velocity area method - rectangular weir - Cippoletti weir - V notch - orifices - Parshall flume - duty of water - irrigation efficiencies - agricultural drainage - surface and sub-surface drainage systems - drainage coefficient

Unit V: Wells and Pumps

Types of wells and sizes – pump types – reciprocating pumps – centrifugal pumps – turbine pumps – submersible pumps – jet pumps.

Practicals

Study of survey instruments - chains and cross staff surveying - linear measurement - plotting and finding areas. Levelling – determination of difference in elevation – Computation of area and volume - Contouring. Problems on Soil loss estimation -USLE - Design of contour bund and graded bund. Farm pond - Components- Designed -Problems on water measurement. Problems on duty of water and irrigation efficiencies. Problems on water requirement and drainage coefficient, Study of different types of wells and its selection. Study on pumps–- Problems on pumps. Visit to soil and water conservation areas.

Theory Schedule

1. Introduction - land surveying - uses in agriculture - Chain cross staff
2. Dumpy level - setting, types of levelling - computation of land slope - difference in elevation.
3. Computation of area and volume – Simpson's rule and Trapezoidal rule.
4. Soil Erosion – causes and evil effects of soil erosion – geologic and accelerated erosion
5. Water erosion - causes - erosivity and erodibility – Stages of water erosion - Splash, sheet, rill and gully erosion - ravines - land slides - Soil loss estimation by universal soil loss equation
6. Wind erosion - factors influencing wind erosion - mechanics of wind erosion – suspension, saltation, surface creep - Effects of water and wind erosion,
7. Erosion control measures for agricultural lands – biological measures – contour cultivation – strip cropping - Cropping systems – vegetative barriers - Windbreaks and shelterbelts - shifting cultivation
8. Mechanical measures – contour bund – graded bund - Broad beds and furrows – basin listing – random tie ridging - Mechanical measures for hill slopes – contour trench – bench terrace – contour stone wall
9. **Mid semester examination.**
10. Rainwater harvesting - Runoff water harvesting - Runoff Computation,
11. Ground water recharge techniques
12. Farm ponds and percolation ponds
13. Irrigation - measurement of flow in open channels - velocity area method - Rectangular weir - Cippoletti weir - V notch - Orifices - Parshall flume
14. Water duty - Irrigation efficiencies
15. Agricultural drainage – need – surface and subsurface drainage systems - drainage coefficient
16. Types of wells and sizes - Suitability
17. Types of pumps and working of – reciprocating pumps -Centrifugal pumps - Turbine pumps – submersible pumps - Jet pumps

Practical schedule

1. Study of survey instruments - chains - compass - plane table - dumpy level.
2. Chains and cross staff surveying - linear measurement - plotting and finding areas.
3. Levelling – determination of difference in elevation - rise and fall method

4. Levelling – determination of difference in elevation -height of collimation method
5. Computation of area – Mid ordinate, average ordinate, Simpson's – Trapezoidal rule
6. Computation of volume – Contouring - Simpson's – Trapezoidal rule
7. Design of contour bund and graded bund.
8. Problems on Soil loss estimation
9. Farm pond Components and Design
10. Problems on water measurement.
11. Problems on duty of water, irrigation efficiencies.
12. Problems on water requirement and drainage coefficient.
13. Study of different types of wells and its selection.
14. Study on pumps
15. Problems on pumps.
16. Visit to soil and water conservation areas.
17. **Final Practical examination.**

References

1. Basak, N.N. 2008. Surveying and Levelling. 25th reprint. Tata Mc-Graw Hill Publishing Company Ltd
2. Michael, A.M. and Ojha, T.P. 2008. Irrigation Theory and Practice. Second Edition. Vikas Publication House, New Delhi

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2. http://soilwater.okstate.edu/courses/lectures-powerpoint

AGR 203 Study Short Tour (0+1)

The students will undertake the short tour during third semester for seven days covering KVK's, Research stations, Sister campuses and ICAR institutes in the southern part of Tamil Nadu. The study tour will provide an exposure to the students to know about the soil, climatic conditions and cropping patterns in the respective agro-climatic zones. The students will also have first-hand information on latest technologies on various crops and allied activities.

NSS 201 National Service Scheme (0+1)

Practicals

Unit I: Popularization of agro techniques and youth on agribusiness

Conducting field days with KVK to popularize improved agro techniques-Conducing seminar / workshop in a nearby village to motivate the youth on agribusiness (involving DEE, KVK, NGO and local agripreneurs).

Unit II: Campaign on self-employment

Campaign on self-employment opportunities like Apiculture, mushroom cultivation, Food processing and value addition, production of biocontrol agents and biofertilizers, nursery techniques, seed production, tissue culture, vermicompost, manufacture of small gadgets and agricultural implements as per local needs and feasibility.

Unit III: Campaign on Veterinary and First aid & Emergency

Animal health care campaign – Dairy and poultry farming - Forage production techniques and silage making - Training the NSS volunteers on road safety measures in involving traffic wardens and RTO. Training NSS volunteers on First AID and emergency call involving NGOs and organizations like St. John's Ambulance, Red Cross, etc.,

Unit IV: Awareness on Road safety and small savings

Organizing Road safety rally- Motivating NSS Volunteers on small savings concept and conveying the message to the public through them.

Unit V: National integration and communal harmony activities

Observation of National integration and communal harmony - Campus development and greening activities

1. **Practical Schedule**
2. Conducting field days with KVK to popularize improved agro techniques.
3. Conducing seminar / workshop in a nearby village to motivate the youth on agribusiness (involving DEE, KVK, NGO and local agripreneurs).
4. Campaign on self-employment opportunities like Apiculture, mushroom cultivation, production of biocontrol agents and biofertilizers.
5. Campaign on self-employment opportunities like Food processing and value addition, nursery techniques, seed production, tissue culture and vermicompost.
6. Campaign on self-employment opportunities manufacture of small gadgets and agricultural implements as per local needs and feasibility.
7. Animal health care campaign – Dairy and poultry farming - Forage production techniques

and silage making.

8. Training the NSS volunteers on road safety measures in involving traffic wardens and RTO.
9. Training NSS volunteers on First AID and emergency call involving NGOs and organizations like St.John’s Ambulance, Red Cross, etc.,
10. **Mid semester Examination**
11. Organizing Road safety rally.
12. Motivating NSS Volunteers on small savings concept and conveying the message to the public through them.
13. Motivating NSS Volunteers on small savings concept and conveying the message to the public through them.
14. Observation of National integration and communal harmony.
15. Observation of National integration and communal harmony
16. 15 -16 Campus development and greening activities
17. **Final Practical Examination.**

References

1. https://nss.gov.in/

NCC 201 National Cadet Corps (NCC) - III (0+1)

Practicals

Unit I: Drill and Types

Drill - Foot Drill (FD) – Drill basics and comments. Drill with Arms (AD) - Attention, stand at ease and stand easy, Getting on parade with rifle and dressing at the order, Dismissing and falling out, Ground/take up arms, resent from the order and vice-versa, Gen Salute, Salami Shastra, Squad Drill, Short/long trail from the order and vice-versa, Examine Arms. Ceremonial Drill (CD) - Guard Mounting, Guard of Honour, Platoon/Coy Drill, Instructional Practice

Unit II: Personality Development and Weapon Training

Introduction to Personality Development. Factors Influencing / shaping Personality: Physical, Social, Psychological and Philosophical. Self-Awareness - Know yourself/ Insight. Change your mind set. Interpersonal relationship and communication. Communication Skills: Group Discussions/ Lecturettes. Hands on training on Weapon handling and Short-Range firing

Unit III: Disaster Management

Civil Defence Organisation, Types of emergencies /Natural Disasters, Fire Services & Fire fighting, Traffic control duringDisaster under Police Supervision, Essential services and their maintenance, Setting up of relief camp, during Disaster Management, Collection & Distribution of Aid material, Assistance during Natural/Other Calamities: Flood /Cyclone /Earthquake /Accident etc

Unit IV: Map Reading

History, Geography & Topography of Border/ Coastal Areas. Cardinal points and Types of North, Types of bearings and use of Service Protractor, Prismatic compass and its use & GPS, Setting a Map, finding North and own position, Map to Ground, Ground to Map, Point to Point March

Unit V: Social Service and Community Development

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

Practical Schedule

1. Foot Drill (FD) – Drill basics, attention, stand at ease, stand easy, comments
2. Arms Drill (AD – I) – Getting on parade with rifle, dressing at the order, dismissing, falling out
3. Arms Drill (AD – II) – Ground/take up arms, present from the order and vice-versa, examine arms
4. Arms Drill (AD – III) – General Salute, Salami Shastra, Squad Drill, short/long trail
5. Ceremonial Drill (CD) – Guard Mounting, Guard of Honour, Platoon/Company Drill, instructional practice
6. Personality Development – Introduction, factors shaping personality (physical, social, psychological, philosophical)
7. Personality Development – Self-awareness, mindset change, interpersonal relations, communication skills (group discussion, lecturettes)
8. Weapon Training – Hands-on weapon handling, safety, and short-range firing practice
9. **Mid Semester Examination**
10. Disaster Management – Civil Defence Organisation, types of emergencies/natural disasters, role of fire services & fire fighting
11. Disaster Management – Traffic control, maintaining essential services, setting up relief camps
12. Disaster Management – Aid collection/distribution, assistance during floods, cyclones, earthquakes, accidents
13. Map Reading – History, geography & topography of border/coastal areas
14. Map Reading – Cardinal points, types of north, bearings, service protractor
15. Map Reading – Prismatic compass, GPS, setting a map, finding north/own position, map-to-ground, ground-to-map, point-to-point march
16. Social Service – Blood Donation Camp, Swachhata Abhiyan, Constitution Day activities -Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao, state/national level community initiatives
17. **Final Practical Examination**

References

1. Handbook of NCC Cadets for 'A', 'B' and 'C' Certificate Examinations,
2. Cadet's Handbook (Army Wing),

3. Gupta, R. K. (2020). *Handbook of NCC Cadets for 'A', 'B' and 'C' Certificate Examinations*. Ramesh Publishing House. ISBN: 978-9387918573.
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2. https://indiancc.mygov.in/
3. https://mybharat.gov.in/Gov/Department/national-cadet-corps

PED 101 Physical Education, First Aid, Yoga Practice and Meditation (0+2)

Practicals

Unit I: Volleyball & Basketball

Volleyball - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques- Basketball - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques - Tennis - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques.

Unit II: Kabaddi & Kho-Kho

Kabaddi - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques – Table Tennis- Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques - Kho-Kho - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques - Physical education.

Unit III: Training and Coaching Methods

Training and Coaching - Meaning and Concept; Methods of Training; aerobic and anaerobic exercises; Calisthenics; Weight training, circuit training, interval training and Fartlek training.

Unit IV: First Aid

First aid related with Skin, Burns, related with Sense organs, Handling and transport of injured traumatized persons. Sports injuries and their treatments.

Unit V: Psychology & Posture

Personality, its dimensions and types; Role of sports in personality development; Motivation and Achievement in Sports; Learning and Theories of learning; Adolescent Problems and its Management; Posture; Postural Deformities; Exercises for good posture.

Practical Schedule

1 to 5	:	Volleyball - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques
6 to 10	:	Basketball - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques
11 to 17	:	Tennis - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques

18 to 20	:	Kabaddi - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques
21 to 25	:	Table Tennis - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques
26 to 30	:	Kho-Kho - Latest rules, measurements of playfield, specifications of equipment and teaching of skills and techniques
31 & 32	:	Physical education; Training and Coaching - Meaning and Concept; Methods of Training; aerobic and anaerobic exercises; Calisthenics; Weight training, circuit training, interval training and Fartlek training; First aid related with Skin, Burns, related with Sense organs, Handling and transport of injured traumatized persons. Sports injuries and their treatments.
33 & 34	:	Personality, its dimensions and types; Role of sports in personality development; Motivation and Achievement in Sports; Learning and Theories of learning; Adolescent Problems and its Management; Posture; Postural Deformities; Exercises for good posture.

References

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3. Roberts, G. C. (Ed.). 2012. *Advances in motivation in sport and exercise*. Human Kinetics. Pages 452.
4. Schunk, D. H. 2020. *Learning theories: An educational perspective*. Pearson. Pages 592.
5. Nicolson, D., & Ayers, H. 2004. *Adolescent problems*. Routledge. Pages 168.
6. Johnson, J. 2015. *Postural correction.* Human Kinetics. Pages 232.
7. Schrag, M. 2018. *The sports rules book.* Human Kinetics. Pages 400.
8. Stubbs, R. (Ed.). 2024. *The Sports Book: The Games, the Rules, the Tactics, the Techniques.* DK Publishing. Pages 456.
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